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Brain Computer Interfaces for Fitter Brains & better Mental health

Humankind is going through a crisis which will change the way we work and live. If the current trends are anything to go by, people will give top priority to health & wellness going forward. A WHO report had earlier indicated that ~20% of the population in India is suffering from mental health issues. Current events will not make it any better.

The importance of mental health as a priority is not just for a general feeling of happiness and wellbeing, but also as a scientifically proven precursor to better physical health and disease prevention. Health insurers and governments will increasingly find that prevention is way cheaper than cure and hence invest in or reimburse such products and services. Addictions and dependencies to opioids, performance enhancement substances, psychedelics might rise for a period before the population realizes that what were sold as magic pills or potions are the new problems they need to wean themselves off.

Advances in neuroscience over the last few decades have been steadily providing people with astounding opportunities in the pursuit of understanding their brain function and enhancing it.

It is important to understand that the mind and brain are not one and the

same. The mind is an intangible manifestation, but the brain is a physical organ that is most responsible for our thoughts, emotions, actions, memory, fears, imagination and more. The brain has around a 100 trillion neuron connections and is constantly generating electrical activity to enable us to perform all physical and 'mental' activities. We can measure and help improve its functioning in a scientific manner.

This involves using a Brain Computer Interface (BCI): a host of sensors, sophisticated hardware and software that connect the brain to a computer so that the electrical activity (brainwaves) can be acquired and analysed.

BCIs provide an evidence-based, risk-free approach, proven to have helped businessmen, athletes, celebrities, students and just about anyone, in enhancing all aspects of their performance. The most advanced ones are developed in the US with support of DARPA (Defence Advanced Research Projects Agency), and the US Department of Defense.

There are two types of BCIs, invasive (which require inserting sensors physically inside the brain) and non-invasive (which need the sensors only to be placed on the scalp). The non-invasive tech could evolve to levels which don't even need to be placed on the head. There is already a booming number of startups with low cost, low efficacy devices for curious users at home promising better meditation or sleep or calmness. A lot of the snake oil will make way for serious tech with real outcomes. The real deep tech needs professionals with expertise and experience of understanding the neurometric data, and using it to assess and enhance client brain functions. Hardware players will then

try and service all sorts of consumers using this technology.

Invasive solutions such as Elon Musk's Neuralink and others which provide deep brain stimulation could prove to be a boon for patients who have severe neurological and mental health disorders, or even those who are paralysed, to be able to communicate, use prosthetics and control external devices. They could also be used by healthy individuals wanting additional computing power attached to their brains.

Growth of BCIs in India does face challenges in terms of access to futuristic materials and talent – requires understanding of data science, neural pathways, nuanced experience of using it with different mental health issues. There are archaic versions which use very few sensors which limit the quality of data and which use game playing which rely on the conscious mind and are no good. Furthermore, like all disruptive technologies in history that have reduced the importance of existing gatekeepers, here too the mainstream medicine thriving on years of psychiatric drugs prescriptions or 'therapy' will continue to resist whilst they can. The wise and prudent though have adopted and are safely enhancing themselves.

For mental health patients, this will be an era where BCI will fill the huge gap between neurology, psychology, psychiatry and pharmacology – limited treatment options, patient experience in terms of adverse side effects, low efficacy and effectiveness of therapy. BCI will also become mainstream for wider, healthy populations from a wellness and disease prevention perspective and empower many to lead healthier, happier lives, with fitter brains. **1**